PQC DAILY STUDY SERIES

Side-Channel Attacks on PQC Implementations

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Understanding Side-Channel Attacks in the PQC Era

As the world transitions to Post-Quantum Cryptography (PQC), the focus is not just on the mathematical robustness of algorithms like ML-KEM and ML-DSA, but also on their physical implementation. Side-channel attacks (SCAs) exploit information leaked from the physical hardware executing cryptographic operations, rather than from theoretical weaknesses in the algorithms themselves. These attacks are a critical threat to the real-world security of PQC.

How do Side-Channel Attacks work?

SCAs monitor physical characteristics of a device to infer secret data. Common side-channels include:

- **Power Consumption:** Different CPU instructions consume slightly different amounts of power. Power analysis attacks (SPA, DPA) can reveal patterns related to secret keys.
- **Electromagnetic (EM) Emissions:** Electronic devices emit EM fields. These emissions can be captured and analyzed to reveal internal operations.
- **Timing Information:** The time taken to perform a cryptographic operation can depend on the input data, including the secret key.
- **Acoustic Cryptanalysis:** The sounds produced by a device's components (like capacitors) can sometimes leak information.

> **Why are PQC algorithms vulnerable?**
>
> Many PQC algorithms, especially lattice-based ones like CRYSTALS-Kyber (ML-KEM) and Dilithium (ML-DSA), rely on complex mathematical operations like polynomial multiplication and rejection sampling. These operations can have data-dependent execution paths or memory access patterns, creating measurable side-channel leakages if not implemented carefully.

Real-World Applicability to Internet Protocols

The security of protocols like TLS 1.3, SSH, and VPNs, which are starting to incorporate PQC, depends entirely on the secure implementation of the underlying algorithms. A side-channel

vulnerability in a server's ML-KEM implementation could allow an attacker to recover the private key by monitoring its power consumption or EM emissions, completely bypassing the quantum-resistant mathematics. This would allow them to decrypt traffic and compromise the security of the entire communication channel.

Countermeasures and Mitigation Strategies

Protecting against SCAs requires a combination of software and hardware techniques:

Technique	Description	Applicability
Constant-Time Implementation	Ensuring that cryptographic operations take the same amount of time, regardless of the input data. This mitigates timing attacks.	Software
Masking	Splitting secret data into multiple random shares. Operations are performed on the shares, making it harder to correlate side-channel information with the original secret.	Software/Hardware
Hiding	Introducing noise or shuffling operations to obscure the correlation between the side-channel leakage and the data being processed.	Hardware
Formal Verification	Using mathematical methods to formally prove that an implementation is resistant to certain classes of side-channel attacks.	Software

Daily Quiz

1. What is the primary difference between a side-channel attack and a traditional cryptanalytic attack?
2. Why are operations like rejection sampling in PQC algorithms a potential source of side-channel leakage?
3. Explain how a constant-time implementation helps mitigate timing attacks.